MA1522 Final Exam Helpsheet

Chapters 1–7 | side 1: Ch. 1–4 · side 2: Ch. 5–7 | 16 MCQ × 2 + 5 written (68) | **EX** = worked pattern

1 Linear systems

- **REF:** zero rows at the bottom, leading entries move strictly right. **RREF:** also every leading entry is 1 and is the only nonzero entry in its column. RREF is unique; REF is not.
- **Row operations:** $R_i \leftrightarrow R_j$, cR_i ($c \neq 0$), $R_i + cR_j$. $2R_2 + R_1$ **is not one operation.**
- Reduce $(A \mid b)$. Last column a pivot column $\Rightarrow$ **inconsistent**. Otherwise: every column of A a pivot column $\Rightarrow$ unique; else one parameter per non-pivot column $\Rightarrow$ infinitely many.
- Homogeneous $Ax = 0$: always consistent; more variables than equations $\Rightarrow$ nontrivial solutions.
- General solution of $Ax = b$ = one particular solution + general solution of $Ax = 0$.

Unknown constants: never divide by an expression that may be 0. Reduce as far as possible, then split into cases at each pivot candidate ($a = 1$, $a = -2$, otherwise). For a square coefficient matrix, $\det \neq 0 \iff$ unique; then test each root of $\det = 0$ separately.

EX $x + y + az = 1$, $x + ay + z = 1$, $ax + y + z = -2$: $\det = -(a-1)^2(a+2)$. $a = 1$: rows clash $\Rightarrow$ none. $a = -2$: rows sum to $0 = 0 \Rightarrow (1 + s, s, s)$.

2 Matrices, inverses, LU

- $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$, $(AB)^{-1} = B^{-1}A^{-1}$, $\left(A^{\mathrm{T}}\right)^{-1} = \left(A^{-1}\right)^{\mathrm{T}}$. $AB \neq BA$ $AB = 0$ does not give $A = 0$ or $B = 0$; no cancellation unless invertible. $(A+B)^2 = A^2 + AB + BA + B^2$.
- Columns of AB are Ab_j; Ax is a linear combination of the columns of A.
- **Inverse:** $(A \mid I) \to \left(I \mid A^{-1}\right)$. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. $A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A)$, $\operatorname{adj}(A) = \left(A_{ji}\right)$ (transpose of the cofactor matrix).
- **Elementary matrix** E = one row operation applied to I; EA performs it on A. Inverses: swap $\leftrightarrow$ itself; $cR_i \leftrightarrow \frac{1}{c}R_i$; $R_i + cR_j \leftrightarrow R_i - cR_j$. If $E_k \cdots E_1 A = I$ then $A = E_1^{-1} \cdots E_k^{-1}$.
- **LU** (no row swaps, only $R_i + cR_j$ with $i > j$): U = the REF reached; L unit lower triangular with $-c$ in entry (i, j). Solve $Ax = b$: $Ly = b$ (forward), then $Ux = y$ (back).

EX $\begin{pmatrix} 2 & 1 & -1 \\ 4 & 5 & -3 \\ -2 & 5 & 3 \end{pmatrix}$: $R_2 - 2R_1$, $R_3 + R_1$, $R_3 - 2R_2 \Rightarrow L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & -1 \\ 0 & 0 & 4 \end{pmatrix}$.

Square A of order n is invertible $\iff$ RREF is $I \iff$ product of elementary matrices $\iff Ax = 0$ only trivially $\iff Ax = b$ unique for all $b \iff \det \neq 0 \iff$ columns (rows) independent / span $\mathbb{R}^n$ / a basis $\iff$ rank $n \iff$ nullity 0 $\iff$ 0 not an eigenvalue $\iff$ has a left (or right) inverse $\iff A^{\mathrm{T}}$ invertible.

2.8 Determinants

- Cofactor expansion along any row or column: $\det = \sum_j a_{ij}A_{ij}$, $A_{ij} = (-1)^{i+j}\det\left(M_{ij}\right)$. Pick the row/column with most zeros. 3×3: $aei + bfg + cdh - ceg - afh - bdi$.
- Triangular $\Rightarrow$ product of the diagonal. Two equal rows / a zero row $\Rightarrow 0$.
- **By reduction:** swap $\Rightarrow \times(-1)$; $cR_i \Rightarrow \times c$; $R_i + cR_j \Rightarrow$ unchanged. So $\det(A) =$ (± product of pivots)/(scalings used).
- $\det(AB) = \det A \det B$, $\det\left(A^{\mathrm{T}}\right) = \det A$, $\det\left(A^{-1}\right) = 1/\det A$, $\det(cA) = c^n \det A$, $\det(\operatorname{adj} A) = (\det A)^{n-1}$. $\det(A + B) \neq \det A + \det B$.
- **Cramer:** $x_i = \det(A_i)/\det(A)$, $A_i = A$ with column i replaced by b.

EX $n \times n$ pattern matrix: expand along row 1 to get a recurrence, e.g. tridiagonal 2, 1: $d_n = 2d_{n-1} - d_{n-2} \Rightarrow d_n = n + 1$.

3 Vector spaces: span, independence, basis

- $v \in \operatorname{span}\{u_1, \ldots, u_k\} \iff (u_1 \cdots u_k \mid v)$ consistent. $\operatorname{span}(S) = \mathbb{R}^n \iff$ REF of $(u_1 \cdots u_k)$ has no zero row. $\operatorname{span}(S_1) \subseteq \operatorname{span}(S_2) \iff$ every vector of S_1 is in $\operatorname{span}(S_2)$.
- **Subspace** = a span = contains $\mathbf{0}$, closed under + and scalar multiples. Solution set of $Ax = 0$: yes. Of $Ax = b \neq 0$: **no** (no $\mathbf{0}$). Products, squares, $\geq$, "or" conditions: usually no — find two vectors whose sum leaves the set.
- **Independent** $\iff c_1u_1 + \cdots + c_ku_k = 0$ only trivially $\iff$ every column of $(u_1 \cdots u_k)$ is a pivot column. More than n vectors in $\mathbb{R}^n \Rightarrow$ dependent. Contains $\mathbf{0} \Rightarrow$ dependent. Two vectors: dependent $\iff$ multiples.
- **Basis** of V: independent and spans V. If $\dim V = k$: any k independent vectors in V form a basis; any k vectors spanning V form a basis. $U \subseteq V \Rightarrow \dim U \leq \dim V$, equal $\iff U = V$.
- **Basis of a solution space:** solve, write $x = sv_1 + tv_2 + \cdots$; the v_i are a basis. Hyperplane $a \cdot x = 0$ in $\mathbb{R}^n$ has dimension $n - 1$.
- **Coordinates:** $[v]_S = (c_1, \ldots, c_k)^{\mathrm{T}}$ with $v = \sum c_iu_i$: solve $(u_1 \cdots u_k \mid v)$.

Transition matrix from T to S: $P = \left([v_1]_S \cdots [v_k]_S\right)$, found from $(S \mid T) \to (I \mid P)$. Then $[w]_S = P[w]_T$; from S to T it is P^{-1}. In $\mathbb{R}^n$: $P = S^{-1}T$ (matrices of basis columns).

EX $\{u, v, w\}$ independent $\Rightarrow \{u + v, v + w, w + u\}$ independent (coefficient matrix has $\det = 2$); $\{u - v, v - w, w - u\}$ dependent (sum is $\mathbf{0}$).

- $\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$.

4 Column space, row space, nullspace, rank

Reduce $A \to R$ (REF or RREF), A is $m \times n$, r = number of pivots.

Space	Basis	dim
$\operatorname{Row}(A)$	nonzero rows of R	r
$\operatorname{Col}(A)$	**columns of A** at the pivot positions of R	r
$\operatorname{Null}(A)$	vectors of the general solution of $Ax = 0$	$n - r$
$\operatorname{Null}\left(A^{\mathrm{T}}\right)$	solve $A^{\mathrm{T}}y = 0$; equals $\operatorname{Col}(A)^{\perp}$	$m - r$

- Row operations preserve Row and Null, and the dependence relations between columns, **not** Col.
- $\operatorname{rank} A = \operatorname{rank} A^{\mathrm{T}} \leq \min(m, n)$; rank + nullity $= n$ (columns). $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$; $\operatorname{rank}\left(A^{\mathrm{T}}A\right) = \operatorname{rank} A$, $\operatorname{Null}\left(A^{\mathrm{T}}A\right) = \operatorname{Null}(A)$.
- $Ax = b$ consistent $\iff b \in \operatorname{Col}(A) \iff \operatorname{rank}(A) = \operatorname{rank}(A \mid b)$. Consistent for **every** $b \iff$ rank $= m$.
- Basis from a spanning set: put the vectors as columns, keep the pivot columns; the RREF entries of a non-pivot column are its coefficients in terms of them. To **extend** to a basis of $\mathbb{R}^n$: add standard basis vectors for the non-pivot columns of the row-form REF.

Full rank. rank $= n$ (columns independent) $\iff A^{\mathrm{T}}A$ invertible $\iff$ left inverse $\left(A^{\mathrm{T}}A\right)^{-1}A^{\mathrm{T}} \iff Ax = b$ has at most one solution $\iff$ least squares solution unique. rank $= m$ (rows independent) $\iff AA^{\mathrm{T}}$ invertible $\iff$ right inverse $A^{\mathrm{T}}\left(AA^{\mathrm{T}}\right)^{-1} \iff Ax = b$ consistent for all b. All right inverses: solve $AX = I$ column by column.

EX Rank with constants c, d: clear column 1; if the remaining rows are $(0, c-4, 0, c-4)$, $(0, 0, d-1, d-1)$ then rank $= 1 + [c \neq 4] + [d \neq 1]$; nullity $= n -$ rank.

✓ MCQ traps

- "$m \times n$, rank $n \Rightarrow$ invertible": **false** unless square.
- A, B invertible says nothing about $A + B$ (take $B = -A$, or $B = A$).
- $\mathbf{0}$ is in every subspace; $\operatorname{Row}(A) \cap \operatorname{Null}(A) = \{\mathbf{0}\}$; $V \cap V^{\perp} = \{\mathbf{0}\}$; $\left(V^{\perp}\right)^{\perp} = V$.
- Row equivalent $\Rightarrow$ same Row, Null, rank; not same Col.
- $A^2 = I$ does not force $A = \pm I$; symmetric × symmetric need not be symmetric (needs $AB = BA$).
- "Statement true for all b / all a": test the extreme cases ($b = 0$, zero matrix, I, $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$).

Proof templates

- **Subspace:** write it as a span or as $\operatorname{Null}(A)$. **Not a subspace:** exhibit vectors breaking closure, or note $\mathbf{0}$ is missing.
- **Independent:** suppose $\sum c_iv_i = 0$, deduce every $c_i = 0$ (apply A, take dot products with each v_j, or compare coordinates).
- **Basis of V:** (1) each vector lies in V; (2) independent; (3) the count equals $\dim V$.
- **Equal subspaces:** $U \subseteq V$ and $\dim U = \dim V$. **Rank:** count pivots, or use rank + nullity = n.
- **Invertible:** produce the inverse, or $\det \neq 0$, or show $Ax = 0 \Rightarrow x = 0$.
- **Norm trick:** $x^{\mathrm{T}}A^{\mathrm{T}}Ax = \|Ax\|^2$, so $A^{\mathrm{T}}Ax = 0 \Rightarrow Ax = 0$.
- **Eigenvalue claims:** start from $Av = \lambda v$, $v \neq 0$, and apply the hypothesis to v.

5.1 Orthogonality & projection

- $\boldsymbol{u}\cdot\boldsymbol{v}=\boldsymbol{u}^{\mathrm{T}}\boldsymbol{v}$; $\|\boldsymbol{u}\|=\sqrt{\boldsymbol{u}\cdot\boldsymbol{u}}$; $d(\boldsymbol{u},\boldsymbol{v})=\|\boldsymbol{u}-\boldsymbol{v}\|$; $\cos\theta=\frac{\boldsymbol{u}\cdot\boldsymbol{v}}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|}$; unit vector $\boldsymbol{u}/\|\boldsymbol{u}\|$. $\|\boldsymbol{u}+\boldsymbol{v}\|^2=\|\boldsymbol{u}\|^2+2\boldsymbol{u}\cdot\boldsymbol{v}+\|\boldsymbol{v}\|^2$.
- Orthogonal set of nonzero vectors $\Rightarrow$ independent. Orthonormal: also all norms 1.
- **Coordinates in an orthogonal basis** (no row reduction): $c_i=\frac{\boldsymbol{w}\cdot\boldsymbol{u}_i}{\boldsymbol{u}_i\cdot\boldsymbol{u}_i}$.
- $V^{\perp}=\{\boldsymbol{w}:\boldsymbol{w}\cdot\boldsymbol{v}=0\text{ for all }\boldsymbol{v}\in V\}$ = Null of the matrix whose **rows** span V. $\dim V+\dim V^{\perp}=n$. $\mathrm{Row}(\boldsymbol{A})^{\perp}=\mathrm{Null}(\boldsymbol{A})$, $\mathrm{Col}(\boldsymbol{A})^{\perp}=\mathrm{Null}\left(\boldsymbol{A}^{\mathrm{T}}\right)$.

Projection of $\boldsymbol{w}$ onto V with **orthogonal** basis $\{\boldsymbol{u}_1,\dots,\boldsymbol{u}_k\}$:

$$\boldsymbol{w}_p=\frac{\boldsymbol{w}\cdot\boldsymbol{u}_1}{\boldsymbol{u}_1\cdot\boldsymbol{u}_1}\boldsymbol{u}_1+\cdots+\frac{\boldsymbol{w}\cdot\boldsymbol{u}_k}{\boldsymbol{u}_k\cdot\boldsymbol{u}_k}\boldsymbol{u}_k.$$

$\boldsymbol{w}-\boldsymbol{w}_p\in V^{\perp}$; $\boldsymbol{w}_p$ is the closest point of V; distance $=\|\boldsymbol{w}-\boldsymbol{w}_p\|$. Basis not orthogonal $\Rightarrow$ Gram–Schmidt first, or solve $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^{\mathrm{T}}\boldsymbol{w}$ and take $\boldsymbol{A}\boldsymbol{x}$.

- Hyperplane $V=\{\boldsymbol{n}\cdot\boldsymbol{x}=0\}$: $\boldsymbol{w}_p=\boldsymbol{w}-\frac{\boldsymbol{w}\cdot\boldsymbol{n}}{\boldsymbol{n}\cdot\boldsymbol{n}}\boldsymbol{n}$; matrix $\boldsymbol{I}-\frac{\boldsymbol{n}\boldsymbol{n}^{\mathrm{T}}}{\boldsymbol{n}^{\mathrm{T}}\boldsymbol{n}}$. Line span$\{\boldsymbol{u}\}$: matrix $\frac{\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}}{\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}}$. Projection matrices: $\boldsymbol{P}^2=\boldsymbol{P}=\boldsymbol{P}^{\mathrm{T}}$, eigenvalues 1 (on V) and 0 (on $V^{\perp}$).
- **Orthogonal matrix** (square): $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}=\boldsymbol{I}\iff\boldsymbol{Q}^{-1}=\boldsymbol{Q}^{\mathrm{T}}\iff$ columns (and rows) orthonormal. $\det=\pm1$, $\|\boldsymbol{Q}\boldsymbol{x}\|=\|\boldsymbol{x}\|$, products of orthogonal matrices are orthogonal. Solve $\boldsymbol{Q}\boldsymbol{x}=\boldsymbol{b}$: $\boldsymbol{x}=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{b}$.

5.4 Gram–Schmidt & QR

$\boldsymbol{v}_1=\boldsymbol{u}_1$, $\boldsymbol{v}_2=\boldsymbol{u}_2-\frac{\boldsymbol{u}_2\cdot\boldsymbol{v}_1}{\boldsymbol{v}_1\cdot\boldsymbol{v}_1}\boldsymbol{v}_1$, $\boldsymbol{v}_3=\boldsymbol{u}_3-\frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_1}{\boldsymbol{v}_1\cdot\boldsymbol{v}_1}\boldsymbol{v}_1-\frac{\boldsymbol{u}_3\cdot\boldsymbol{v}_2}{\boldsymbol{v}_2\cdot\boldsymbol{v}_2}\boldsymbol{v}_2$, …; then $\boldsymbol{q}_i=\boldsymbol{v}_i/\|\boldsymbol{v}_i\|$. Rescale any $\boldsymbol{v}_i$ to clear fractions before continuing (project onto the **new** $\boldsymbol{v}$'s, never the old $\boldsymbol{u}$'s).

QR ($\boldsymbol{A}$ with independent columns): $\boldsymbol{Q}=(\boldsymbol{q}_1\cdots\boldsymbol{q}_n)$, $\boldsymbol{R}=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{A}$, upper triangular with $r_{ij}=\boldsymbol{q}_i\cdot\boldsymbol{a}_j$ and $r_{ii}=\|\boldsymbol{v}_i\|>0$. Check: $\boldsymbol{Q}\boldsymbol{R}=\boldsymbol{A}$.

EX $\boldsymbol{A}=\begin{pmatrix}1&-1\\2&4\\2&1\end{pmatrix}$: $\boldsymbol{q}_1=\frac{1}{3}(1,2,2)^{\mathrm{T}}$, $\boldsymbol{a}_2\cdot\boldsymbol{q}_1=3$, $\boldsymbol{a}_2-3\boldsymbol{q}_1=(-2,2,-1)^{\mathrm{T}}\Rightarrow\boldsymbol{q}_2=\frac{1}{3}(-2,2,-1)^{\mathrm{T}}$, $\boldsymbol{R}=\begin{pmatrix}3&3\\0&3\end{pmatrix}$.

5.5 Least squares

$\hat{\boldsymbol{x}}$ least squares solution of $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}\iff\boxed{\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\hat{\boldsymbol{x}}=\boldsymbol{A}^{\mathrm{T}}\boldsymbol{b}}\iff\boldsymbol{A}\hat{\boldsymbol{x}}=$ projection of $\boldsymbol{b}$ onto $\mathrm{Col}(\boldsymbol{A})$.

- Always consistent. Unique $\iff$ columns of $\boldsymbol{A}$ independent; then $\hat{\boldsymbol{x}}=\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\right)^{-1}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{b}$, or with QR: $\boldsymbol{R}\hat{\boldsymbol{x}}=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{b}$ (back-substitute).
- Columns dependent $\Rightarrow$ infinitely many $\hat{\boldsymbol{x}}=$ particular + Null($\boldsymbol{A}$), but $\boldsymbol{A}\hat{\boldsymbol{x}}$ is the same for all. Error $=\|\boldsymbol{b}-\boldsymbol{A}\hat{\boldsymbol{x}}\|$.
- **Fitting** $y=a+bx$ to (x_i,y_i): rows $(1,x_i)$, unknown $(a,b)^{\mathrm{T}}$, right side (y_i). Normal equations: $\begin{pmatrix}n&\sum x_i\\\sum x_i&\sum x_i^2\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix}=\begin{pmatrix}\sum y_i\\\sum x_iy_i\end{pmatrix}$. Quadratic: rows $\left(1,x_i,x_i^2\right)$. A polynomial of degree $\le n-1$ passes exactly through n points with distinct x_i.

EX $(-1,0),(0,1),(1,3),(2,4)$: $4a+2b=8$, $2a+6b=11\Rightarrow y=1.3+1.4x$.

6.1 Eigenvalues & diagonalization

- $\boldsymbol{A}\boldsymbol{v}=\lambda\boldsymbol{v}$, $\boldsymbol{v}\neq\boldsymbol{0}$. Characteristic polynomial $\det(x\boldsymbol{I}-\boldsymbol{A})$; eigenspace $E_\lambda=\mathrm{Null}(\lambda\boldsymbol{I}-\boldsymbol{A})$. Triangular $\Rightarrow$ eigenvalues on the diagonal. 2×2: $x^2-(a+d)x+\det$.
- $\sum\lambda_i=$ trace, $\prod\lambda_i=\det$. $\boldsymbol{A}^k$: λ^k; $\boldsymbol{A}^{-1}$: $1/\lambda$; $\boldsymbol{A}+c\boldsymbol{I}$: $\lambda+c$; $p(\boldsymbol{A})$: $p(\lambda)$ — same eigenvectors. $\boldsymbol{A}^{\mathrm{T}}$: same eigenvalues. $\boldsymbol{A}^2=\boldsymbol{A}\Rightarrow\lambda\in\{0,1\}$; $\boldsymbol{A}^k=\boldsymbol{0}\Rightarrow\lambda=0$.
- $1\le$ geometric mult. $=\dim E_\lambda=n-\mathrm{rank}(\lambda\boldsymbol{I}-\boldsymbol{A})\le$ algebraic mult.

Diagonalizable $\iff n$ independent eigenvectors $\iff$ char. polynomial splits and geometric = algebraic for **every** eigenvalue. n distinct eigenvalues $\Rightarrow$ yes. Then $\boldsymbol{P}=(\boldsymbol{v}_1\cdots\boldsymbol{v}_n)$, $\boldsymbol{D}=\mathrm{diag}(\lambda_1,\dots,\lambda_n)$ **in the same order**, $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$, $\boldsymbol{A}^k=\boldsymbol{P}\boldsymbol{D}^k\boldsymbol{P}^{-1}$.

- Only repeated eigenvalues need checking. Not diagonalizable: $\begin{pmatrix}1&1\\0&1\end{pmatrix}$, $\begin{pmatrix}0&1\\0&0\end{pmatrix}$. Diagonalizable does not mean invertible, nor conversely.

EX $\begin{pmatrix}1&a&b\\0&2&c\\0&0&1\end{pmatrix}$: $\lambda=1$ twice; $\boldsymbol{C}-\boldsymbol{I}=\begin{pmatrix}0&a&b\\0&1&c\\0&0&0\end{pmatrix}$ needs rank $1\iff b=ac$.

EX $\boldsymbol{A}=\begin{pmatrix}4&-2\\1&1\end{pmatrix}$: $\lambda=2,3$; $\boldsymbol{P}=\begin{pmatrix}1&2\\1&1\end{pmatrix}$; $\boldsymbol{A}^n=\boldsymbol{P}\begin{pmatrix}2^n&0\\0&3^n\end{pmatrix}\boldsymbol{P}^{-1}$. For $\boldsymbol{x}_n=\boldsymbol{A}^n\boldsymbol{x}_0$: write $\boldsymbol{x}_0=c_1\boldsymbol{v}_1+c_2\boldsymbol{v}_2$, then $\boldsymbol{x}_n=c_1\lambda_1^n\boldsymbol{v}_1+c_2\lambda_2^n\boldsymbol{v}_2$.

- $|\lambda|<1$ terms die as $n\to\infty$; the $\lambda=1$ component survives.

6.3 Orthogonal diagonalization

$\boldsymbol{A}$ symmetric $\iff\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{\mathrm{T}}$ with $\boldsymbol{P}$ orthogonal (spectral theorem). Symmetric $\Rightarrow$ always diagonalizable, real eigenvalues, eigenspaces of different eigenvalues orthogonal.

- Method: find eigenspaces; **Gram–Schmidt inside each repeated eigenspace** normalise every vector; columns of $\boldsymbol{P}$.
- Diagonalizable does not imply symmetric; orthogonally diagonalizable does.

EX $\begin{pmatrix}3&1&1\\1&3&1\\1&1&3\end{pmatrix}$: $\lambda=5$: $(1,1,1)/\sqrt{3}$; $\lambda=2$ (plane $x+y+z=0$): $(1,-1,0)/\sqrt{2}$, $(1,1,-2)/\sqrt{6}$. Rebuild from eigen-data: $\boldsymbol{A}=\sum\lambda_i\boldsymbol{q}_i\boldsymbol{q}_i^{\mathrm{T}}$.

6.4 Markov chains

- Stochastic $\boldsymbol{P}$: entries ≥0, **columns sum to 1** entry (i,j) = probability of going **from** j **to** i. $\boldsymbol{x}_n=\boldsymbol{P}^n\boldsymbol{x}_0$; two-step probabilities are entries of $\boldsymbol{P}^2$.
- **Equilibrium:** solve $(\boldsymbol{P}-\boldsymbol{I})\boldsymbol{x}=\boldsymbol{0}$, then scale so the entries sum to 1. 1 is always an eigenvalue. If the chain converges, the limit is an equilibrium vector.

EX $\begin{pmatrix}0.7&0.2\\0.3&0.8\end{pmatrix}$: $0.3x_1=0.2x_2\Rightarrow(0.4,0.6)^{\mathrm{T}}$. 2×2 general $\begin{pmatrix}1-a&b\\a&1-b\end{pmatrix}$: equilibrium $\frac{1}{a+b}(b,a)^{\mathrm{T}}$, other eigenvalue $1-a-b$.

6.5 Singular value decomposition

$\boldsymbol{A}=\boldsymbol{U}\Sigma\boldsymbol{V}^{\mathrm{T}}$, $\boldsymbol{A}$ is $m\times n$, $\boldsymbol{U}$ $m\times m$ and $\boldsymbol{V}$ $n\times n$ orthogonal, Σ $m\times n$ with $\sigma_1\ge\cdots\ge\sigma_r>0$ on the diagonal.

1. $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ ($n\times n$): eigenvalues $\mu_1\ge\cdots\ge\mu_n\ge0$; $\sigma_i=\sqrt{\mu_i}$.
2. $\boldsymbol{V}$: orthonormal eigenvectors $\boldsymbol{v}_i$ of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$, **same order**.
3. $\boldsymbol{u}_i=\frac{1}{\sigma_i}\boldsymbol{A}\boldsymbol{v}_i$ for $i\le r$.
4. Extend to an orthonormal basis of $\mathbb{R}^m$: $\boldsymbol{u}_{r+1},\dots$ from $\mathrm{Null}\left(\boldsymbol{A}^{\mathrm{T}}\right)$ (solve $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{y}=\boldsymbol{0}$, Gram–Schmidt, normalise).

- Wide matrix ($m<n$): do the SVD of $\boldsymbol{A}^{\mathrm{T}}$ and transpose: $\boldsymbol{A}^{\mathrm{T}}=\boldsymbol{V}\Sigma^{\mathrm{T}}\boldsymbol{U}^{\mathrm{T}}$.
- $r=\mathrm{rank}\,\boldsymbol{A}=$ number of nonzero σ_i. $\mathrm{Col}(\boldsymbol{A})=\mathrm{span}\{\boldsymbol{u}_1,\dots,\boldsymbol{u}_r\}$; $\mathrm{Null}\left(\boldsymbol{A}^{\mathrm{T}}\right)=\mathrm{span}\{\boldsymbol{u}_{r+1},\dots,\boldsymbol{u}_m\}$; $\mathrm{Row}(\boldsymbol{A})=\mathrm{span}\{\boldsymbol{v}_1,\dots,\boldsymbol{v}_r\}$; $\mathrm{Null}(\boldsymbol{A})=\mathrm{span}\{\boldsymbol{v}_{r+1},\dots,\boldsymbol{v}_n\}$. $\boldsymbol{A}=\sum_{i\le r}\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^{\mathrm{T}}$.
- Projection of $\boldsymbol{w}$ onto $\mathrm{Col}(\boldsymbol{A})$: $\sum_{i\le r}(\boldsymbol{w}\cdot\boldsymbol{u}_i)\boldsymbol{u}_i$. Eigenvalues of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$: σ_i^2 and zeros.

EX $\begin{pmatrix}2&1\\1&2\\1&-1\end{pmatrix}$: $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}=\begin{pmatrix}6&3\\3&6\end{pmatrix}$, $\mu=9,3$; $\boldsymbol{v}=(1,\pm1)/\sqrt{2}$; $\boldsymbol{u}_1=(1,1,0)/\sqrt{2}$, $\boldsymbol{u}_2=(1,-1,2)/\sqrt{6}$, $\boldsymbol{u}_3=(1,-1,-1)/\sqrt{3}$.

7 Linear transformations

- Linear $\iff T(a\boldsymbol{u}+b\boldsymbol{v})=aT(\boldsymbol{u})+bT(\boldsymbol{v})$. Quick fails: $T(\boldsymbol{0})\neq\boldsymbol{0}$; constants added; products or powers of coordinates.
- **Standard matrix** $\boldsymbol{A}=(T(\boldsymbol{e}_1)\cdots T(\boldsymbol{e}_n))$, $T(\boldsymbol{x})=\boldsymbol{A}\boldsymbol{x}$. Composition $S\circ T\leftrightarrow\boldsymbol{B}\boldsymbol{A}$ (right-most acts first).

$[T]_S=(T(\boldsymbol{u}_1)\cdots T(\boldsymbol{u}_n))$ for a basis S; $T(\boldsymbol{v})=[T]_S[\boldsymbol{v}]_S$; standard matrix $\boldsymbol{A}=[T]_S\boldsymbol{P}$, where $\boldsymbol{P}=(\boldsymbol{u}_1\cdots\boldsymbol{u}_n)^{-1}$ is the transition matrix from E to S. A linear map is determined by its values on a basis (or any spanning set).

- Range $R(T)=\mathrm{Col}(\boldsymbol{A})$, $\mathrm{rank}(T)=\mathrm{rank}\,\boldsymbol{A}$; $\ker(T)=\mathrm{Null}(\boldsymbol{A})$, $\mathrm{nullity}(T)$. rank + nullity $=n$.
- Injective $\iff\ker=\{\boldsymbol{0}\}\iff$ rank n. Surjective $\iff$ rank m. $n>m\Rightarrow$ not injective; $n<m\Rightarrow$ not surjective. $\ker(T)$ nontrivial for square $\boldsymbol{A}\iff\det\boldsymbol{A}=0$.
- Rotation by θ: $\begin{pmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{pmatrix}$. Reflection about span$\{\boldsymbol{u}\}$: $2\frac{\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}}{\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}}-\boldsymbol{I}$. Projection: symmetric, $\boldsymbol{P}^2=\boldsymbol{P}$.
- $\lim T^n(\boldsymbol{x})$: expand $\boldsymbol{x}$ in an eigenbasis; keep the $\lambda=1$ part.

✓ Section B checklist

- Label every row operation; state the pivot / non-pivot columns you read off; define parameters $s,t\in\mathbb{R}$.
- **Checks that cost seconds:** $\boldsymbol{A}\boldsymbol{A}^{-1}=\boldsymbol{I}$; $\boldsymbol{A}\boldsymbol{v}=\lambda\boldsymbol{v}$ for each eigenvector; trace $=\sum\lambda_i$; $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}=\boldsymbol{I}$ and $\boldsymbol{Q}\boldsymbol{R}=\boldsymbol{A}$; $\boldsymbol{A}\boldsymbol{v}_i=\sigma_i\boldsymbol{u}_i$; equilibrium entries sum to 1; least squares residual $\boldsymbol{b}-\boldsymbol{A}\hat{\boldsymbol{x}}$ is $\perp$ to every column of $\boldsymbol{A}$; projection lies in V.
- Parameter questions: answer for **all** values — list each special case and the generic case.
- "Justify": name the theorem (rank–nullity, spectral theorem, invertibility equivalences) or give a counterexample.